Sri Lanka Institute of Information Technology



Green DevOps - Energy efficient CI/CD Pipieline

R26-SE-027

COMMENTS FOR PROPOSAL

Student ID	Pa
IT22110084 (Release Component)	Impro C
IT22149930 (Deployment Component)	C

CURRENT PROGRESS AND NEXT PHASES

	PP1	
Release Component	Change Impact Analyzer, Selective Build Execution, Selective Test Execution done.	emi bu
Deploy Component	The energy profiler & deployment comparator Implemented	I decis with
Operation	Future workload prediction	Ieh

WHY OUR SOLUTION IS VALUABLE

The Real-World Problem

- Rapid growth of cloud-native systems and CI/CD pipelines creates green debt.
- DevOps pipelines prioritize speed over energy efficiency.
- Small code changes trigger full builds and tests.
- This causes resource waste, high energy usage, and increased carbon emissions.

Carbon emissions impact the whole society through

- climate change
- extreme weather
- health risks
- rising sea levels
- and damage to ecosystems

Carbon Footprint

ICT sector produces **2–4%** of global CO_2 emissions.

Resource Waste

About **90%** CPU time wasted during full rebuilds.

Economic Impact

35% annual increase in cloud costs.

Our Environment Setup

Oracle Jump Server + Reverse SSH Tunnel Setup

1) Admin PC / Client



Admin PC

ssh -p 2510 <user>@147.14.144.192

1) SSH to
147.14.144.192:2510

Internet

Oracle Cloud Free Tier Instance

Jump Server / Bastion Host

Static Public IP

Publicly Reachable from Internet

Port 22: SSH service

Port 2510: Exposed reverse tunnel endpoint

Internal Mapping (Reverse Tunnel)

2510 -> Ubuntu Laptop:22

External clients connect to 147.14.144.192:2510
Traffic is forwarded through the reverse SSH tunnel to the Lubuntu Laptop on port 22.

2) Reverse SSH tunnel from Lubuntu -> Oracle

3) Home / Local Network



Lubuntu Laptop
Local Server

Runs SSH server on port 22

Behind NAT / no public IP

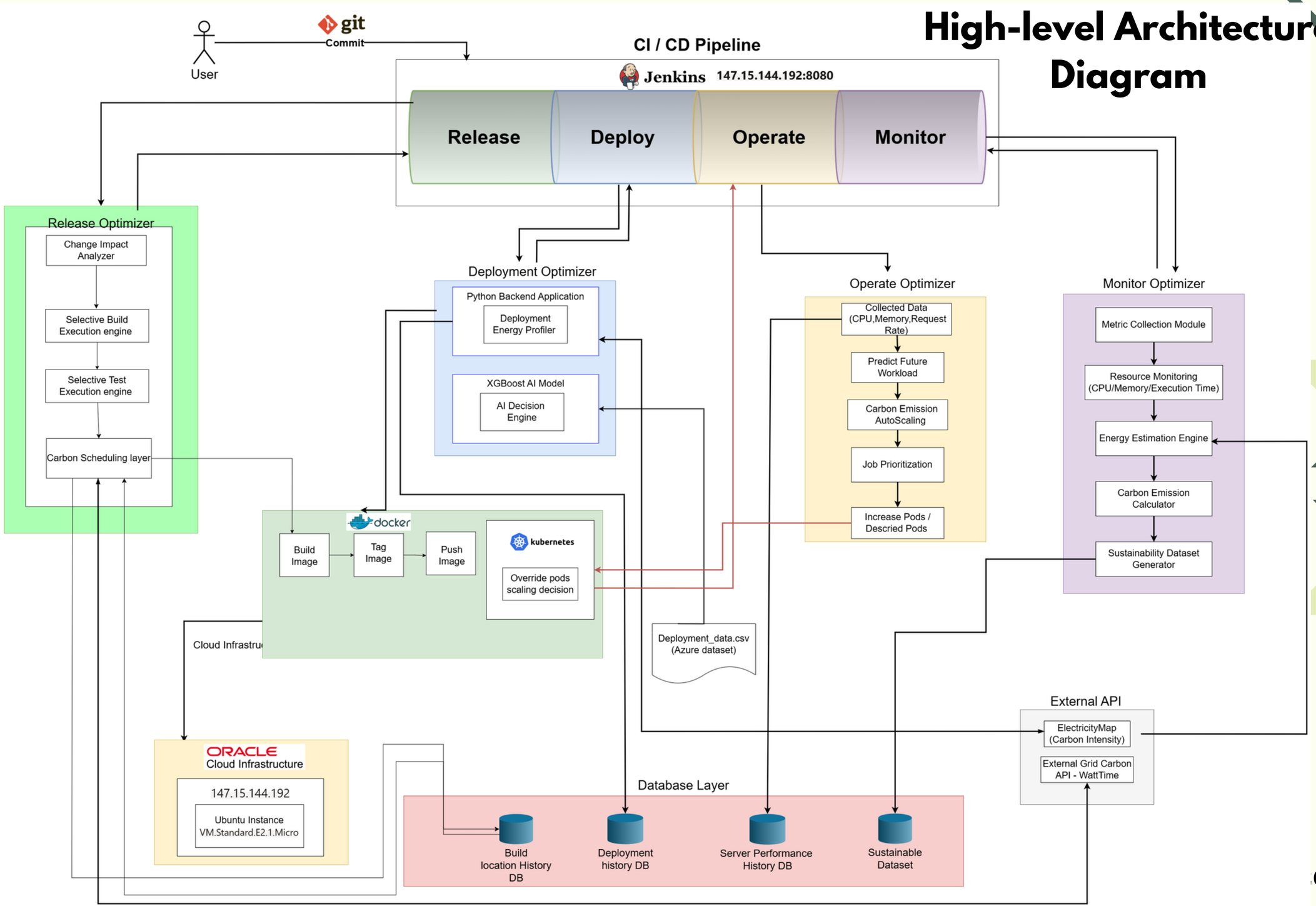
Maintains persistent reverse SSH tunnel

autossh -N -R *:2510:localhost:22
opc@147.14.144.192

Admin PC → 147.14.144.192:2510 → Reverse SSH tunnel → Lubuntu Laptop:22

End-to-end path from Admin PC to Lubuntu Laptop via Oracle Jump Server

High-level Architecture Diagram





IT22110084 | Mendis J.D.L.

Release stage Component

Specialization: Software Engineering

Release stage Component

Problem in Traditional CI/CD

Traditional pipelines typically perform

- Full system builds for every commit
- Execution of the entire test suite
- Builds unaware of green energy.

Leads to

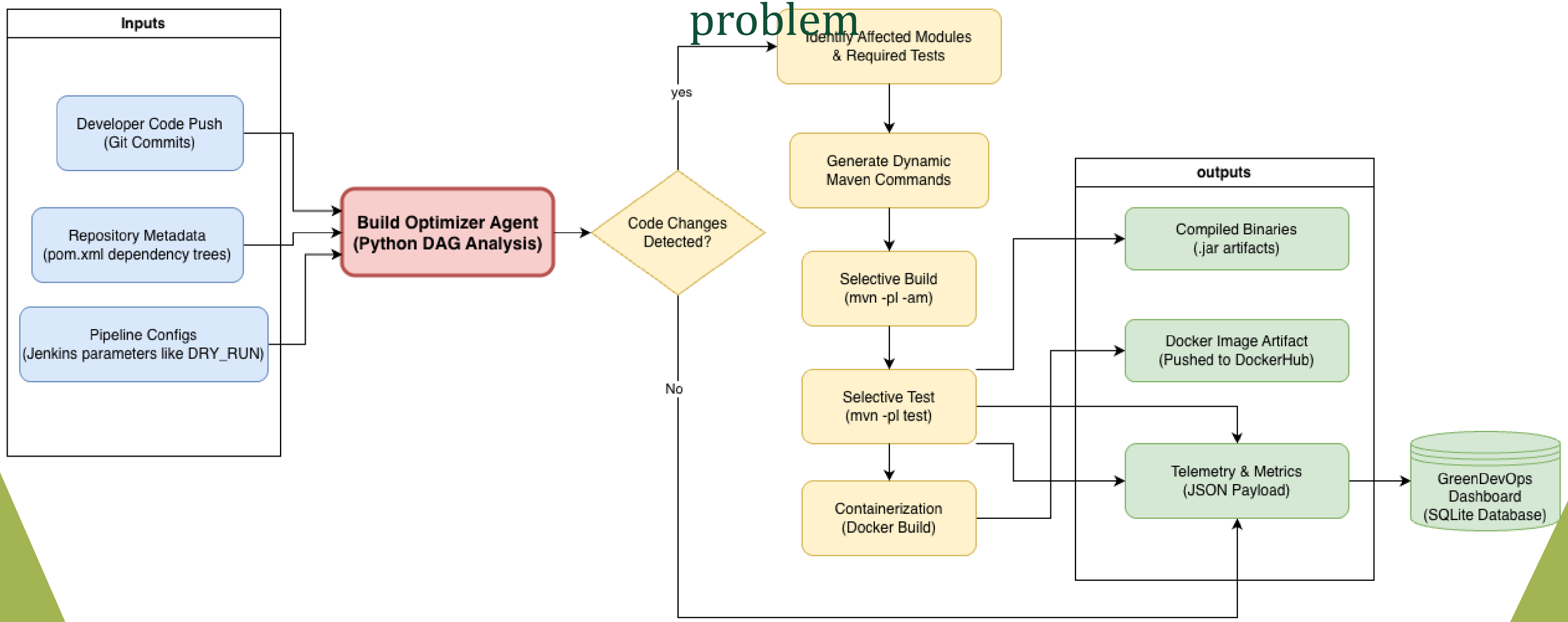
- High computational overhead
- Increased energy consumption
- Higher carbon emissions from cloud infrastructure

Proposed Solution :

- **Plugin which**
  - performs Impact analysis of commits
  - performs Selective builds and test execution
  - Suggests build times and regions aware of green energy

Release stage Component

How the solution will address the problem



Release stage Component

What I implemented

Jenkins-based release pipeline

- **Methodology: CI/CD pipeline automation**
- **Technologies: Jenkins Pipeline, Groovy, Maven, GitHub**

Selective build and test execution

- **Methodology: incremental build and test reduction**
- **Technologies: Maven -pl, Maven -am, Surefire test reports, Jenkins stages**

Impact Analyzer for changed Maven module detection

- **Methodology: change impact analysis and selective release optimization**
- **Technologies: Git diff, Python build optimizer, Maven multi-module project structure**

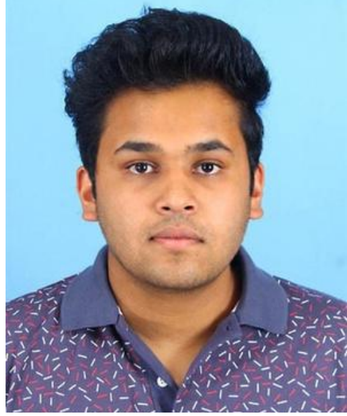
Release metrics sent to GreenDevOps dashboard

- **Methodology: CI/CD observability and measurement**
- **Technologies: REST API, JSON payloads, Flask dashboard, SQLite database**





Maven™



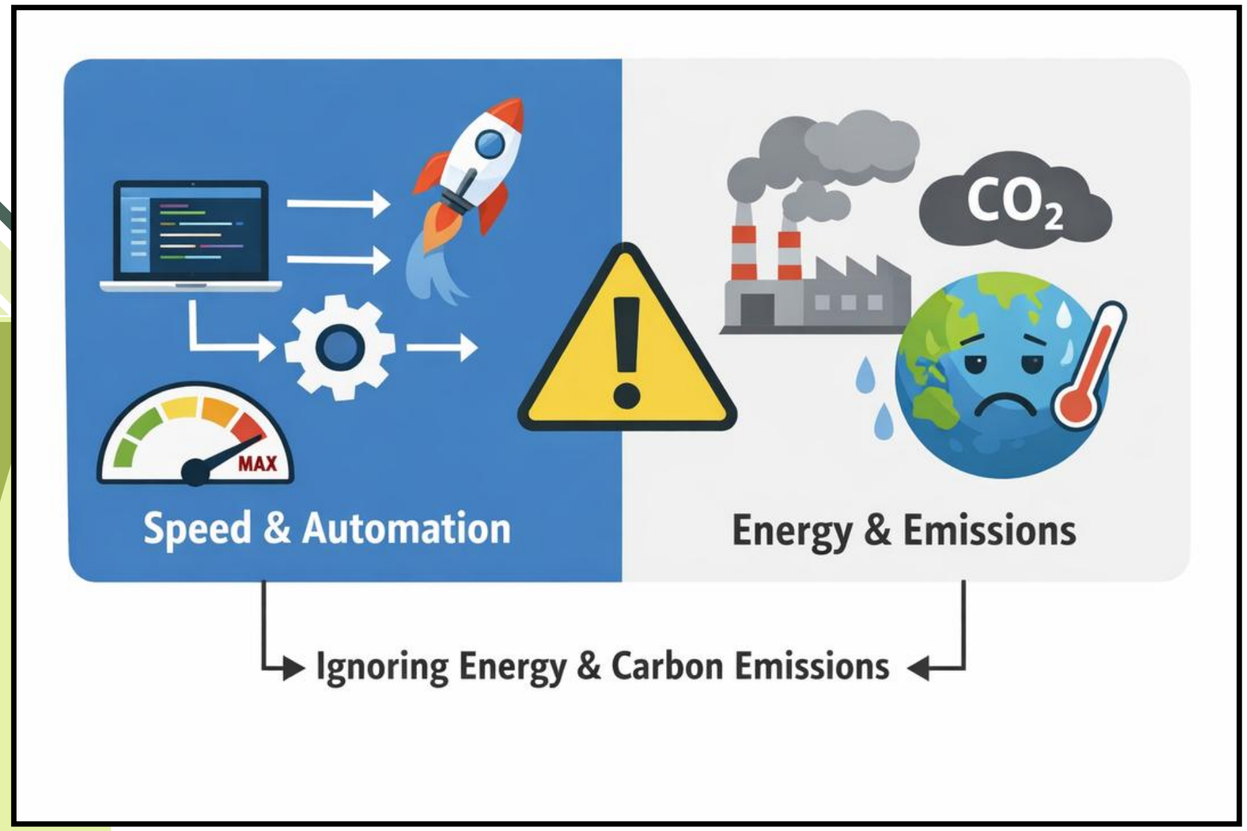
IT22149930 | Rathnayake D.M.B.H.

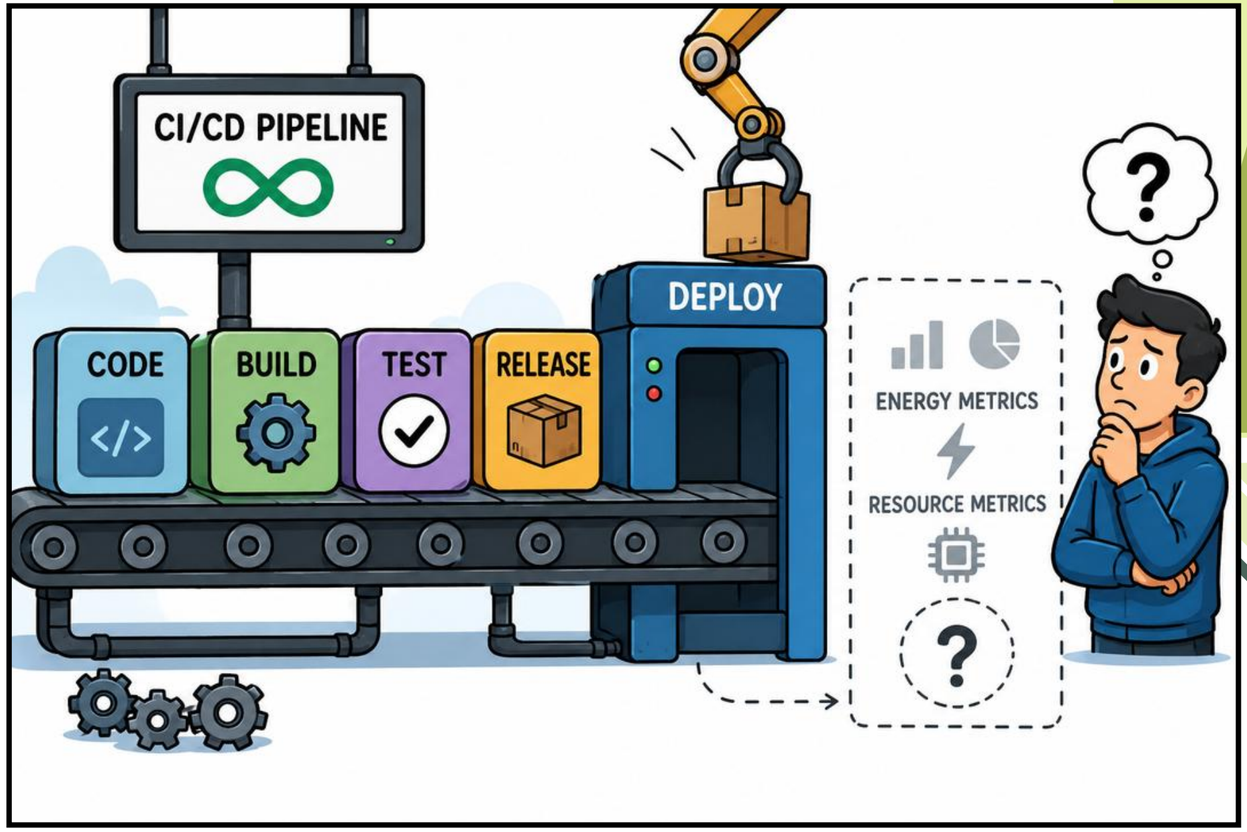
Deploy Stage Component

Specialization: Software Engineering

Deploy Stage Component

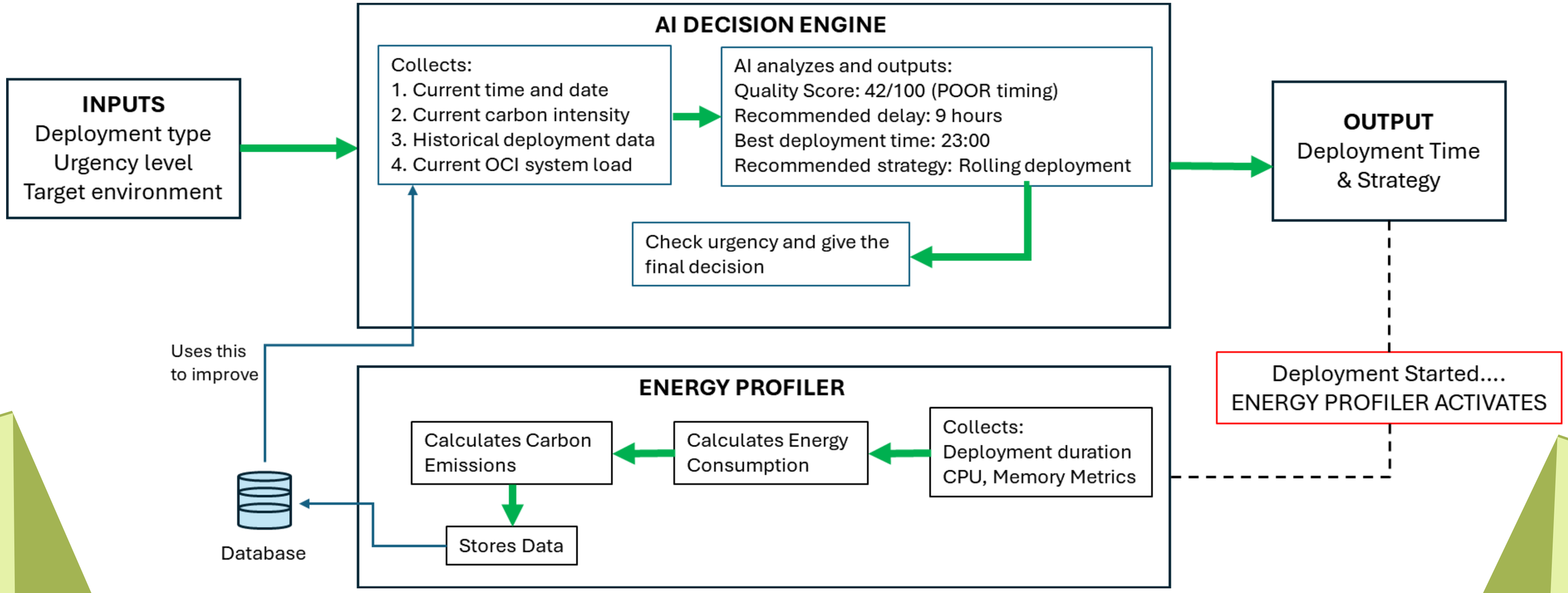
Problem





Deploy Stage Component

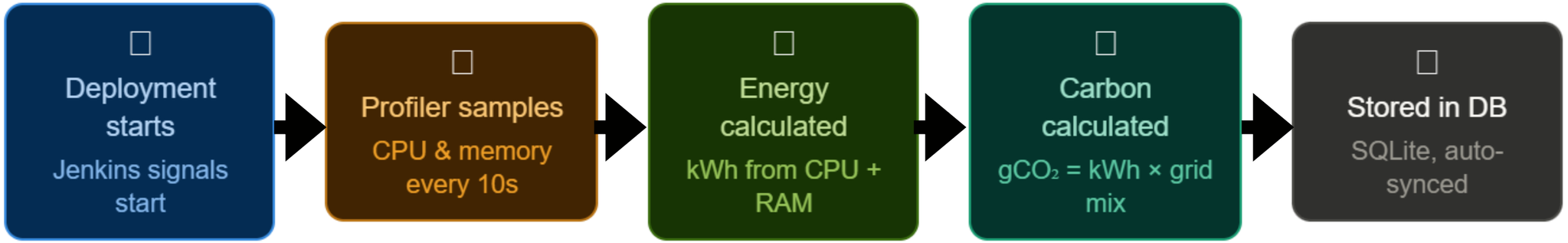
How the Solution will address the Problem



Deploy Stage Component

Every deployment burns energy. Nobody is tracking the carbon cost.

I built a system that watches a deployment happen, measures how hard the server works, and converts that into a real CO_2 number.



profiler.py — Measures CPU & memory
energy_calculator.py — Converts usage to kWh
carbon_api.py — Fetches live grid intensity
carbon_calculator.py — Turns kWh into CO_2
db_sync.py — Saves everything to database



Operation Stage Component

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Specialization: Software Engineering



Operation Stage Component

Sub-Problem

•Traditional DevOps scaling is reactive
•No consideration of Energy consumption and Carbon emissions
•Leads to:
- Resource wastage
- High operational cost
- Environmental impact



Resource waste and
higher environmental impact

User Requirements Addressed

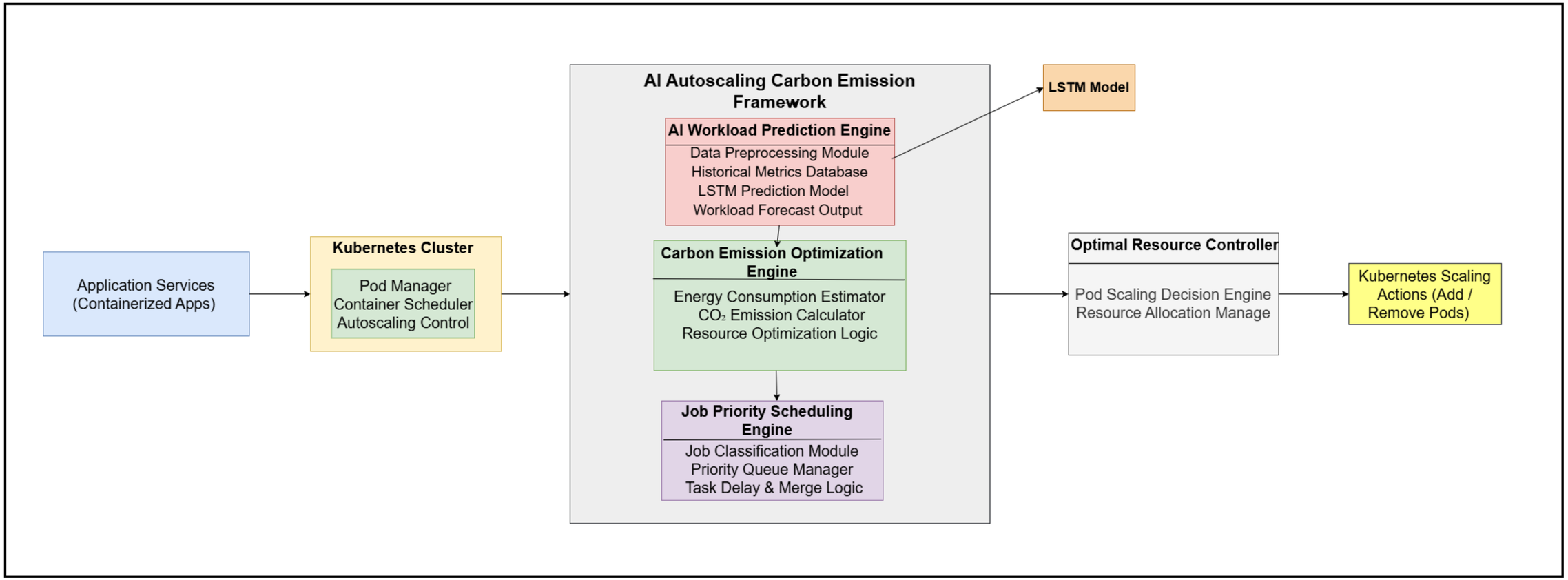
- Efficient resource utilization
- Reduced cloud cost
- Lower carbon emissions
- Maintain SLA performance
- Smart autoscaling decisions

Proposed Solution

- AI Workload Prediction Engine
- Carbon Optimization Engine
- Job Prioritization Engine

Operation Stage Component

Proposed System Architecture



Tool And Tecnology

kubernetes

ORACLE Cloud Infrastructure

python™

Long / Short Term Memory (LSTM)

Prometheus



IT22189776 | Kodithuwakku I.P.

Monitor stage Component

Specialization: Software Engineering

Monitor Stage Component

Problem

Traditional CI/CD pipelines focus on speed and automation but do not provide visibility into their environmental impact.

Leads to

- Unmeasured energy consumption
- Hidden carbon emissions
- Difficulty identifying abnormal behavior
- Lack of data-driven sustainability insights

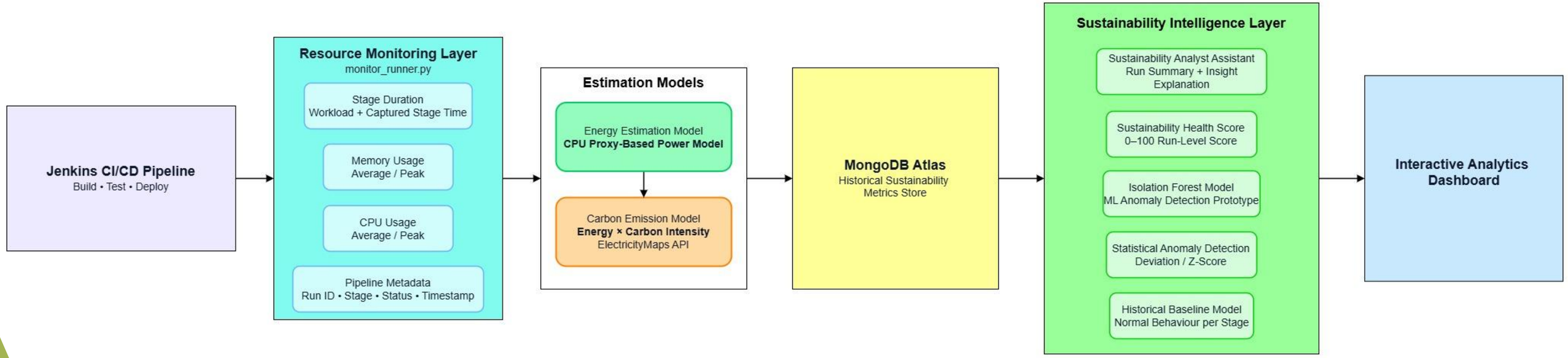
Proposed Solution

Develop a Sustainability Framework that:

- Monitors stages Individually
- Estimates energy consumption and carbon emissions
- Detects abnormal behavior
- Generates insights through the Sustainability Analyst Assistant
- Visualizes historical and stage-level analytics in an interactive dashboard

Monitor Stage Component

How the solution will address the problem



THANK YOU

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